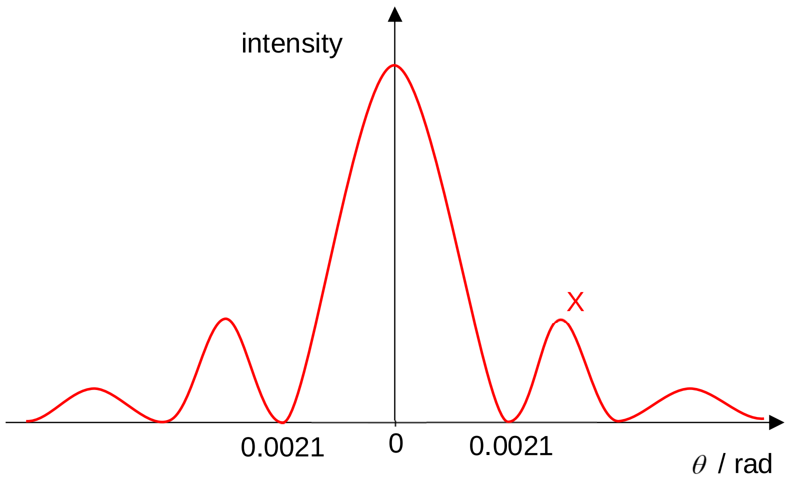
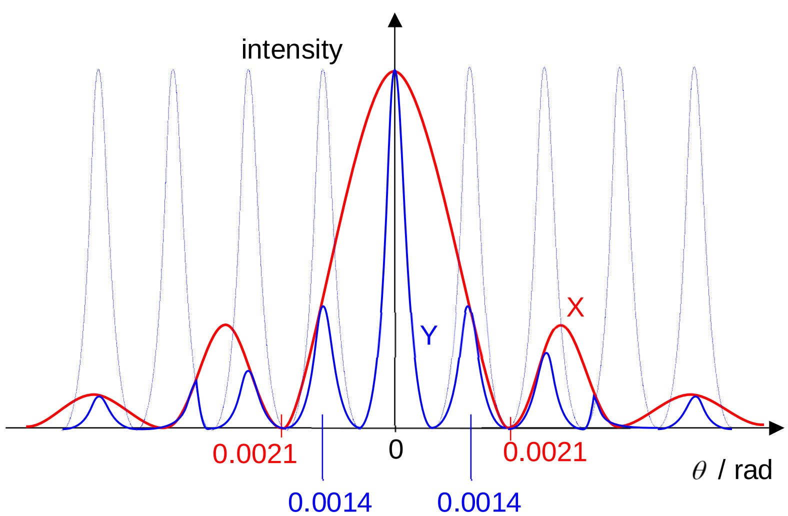
	= 313 m s	
9(a)(i)	spreading of waves into region of geometric shadow as they pass through an aperture or past an obstacle	**B1**
9(a)(ii)	when two or more waves of the same type meet and overlap, resultant displacement at any point is the vector sum of the individual displacements	**B1**
9(b)(i)	$\sin\theta = \dfrac{\lambda}{b}$ $\theta = \sin^{-1}\left(\dfrac{\lambda}{b}\right) = \sin^{-1}\left(\dfrac{630\times10^{-9}}{0.30\times10^{-3}}\right)$ $= 0.0021 \text{ rad}$	**C1** (substitution) **A1**
9(b)(ii)		1st angle labelled allow ecf from **(b)(i)** symmetric at least 4 minima

Qns	Answer	Marks
9(b)(iii)	$\tan\theta = \dfrac{\text{fringe to fringe distance}}{\text{screen distance}} = \dfrac{x}{D}$ $x = \dfrac{\lambda D}{a}$ $\dfrac{x}{D} = \dfrac{\lambda}{a}$ $\tan\theta = \dfrac{\lambda}{a}$ $\theta = \tan^{-1}\left(\dfrac{\lambda}{a}\right) = \tan^{-1}\left(\dfrac{630\times10^{-9}}{0.45\times10^{-3}}\right)$ $= 0.0014$ rad  (dotted: double slit pattern without considering single slit envelope)	1[st] fringe labelled no ecf symmetric at least 3 maximas within central single slit width
9(b)(iv)1.	fringe separation increase	B1
9(b)(iv)2.	screen generally illuminated with no observable pattern	B1
9(c)	$d\ \sin\theta = n\lambda$ $\theta \to 90°$ $n \le \dfrac{d}{\lambda} = \dfrac{10^{-3}}{(300)(630\times10^{-9})} = 5.29$ maximum visible order is 5 so <u>11 bright spots</u> including central max Comments: Many candidates wrote the answer as 5 and forgot to multiply it by 2 (5 spots at each side of maxima) and adding 1 more (central maxima).	**C1** (substitution) **A1**
9(d)(i)	thermal energy required per unit mass to change the phase of a substance from solid phase to liquid phase at constant temperature.	**B1**

Qns	Answer	Marks
9(d)(ii)	thermal energy required per unit mass to raise the temperature of a substance by one degree kelvin	B1
9(e)	thermal energy to bring ice to 0°C $= mc_{ice}\Delta T$ $= (24\times10^{-3})(2.1\times10^{3})(15)$ $= 756$ J thermal energy to melt ice to 0°C $= mL_{ice}$ $= (24\times10^{-3})(3.3\times10^{5})$ $= 7920$ J let final temperature $= T_f$ thermal energy to bring ice to T_f = energy loss from 200 g water to T_f $756 + 7920 + m_{melt}c_w(T_f - 0) = m_{200g}c_w(28 - T_f)$ $8676 + (24\times10^{-3})(4.2\times10^{3})(T_f) = (200\times10^{-3})(4.2\times10^{3})(28 - T_f)$ $T_f = 15.8$ °C	**C1** (either 756 J or 7920 J) **C1** heat capacity equations relating two processes to final temperature **A1**
9(f)	underestimate beaker supplys heat to melt and warm up ice so less-than-expected thermal energy is lost from 200 g of water final temperature likely higher than calculated	**B1** **B1** **A0**
9(g)	(table below)	**B1 each row**

	ΔU	Q	W
an ideal gas is compressed in an insulated container	+	0	+
a solid is cooled without a change in volume	-	-	0
water is boiling at 100°C	+	+	-